

Feature Engineering (FE)

Feature engineering is applicable for **Supervised Learning** only.

FE. is applied on INDEPENDENT VARIABLES.
(feature cols)

Goal:

The ideal goal of FE is to

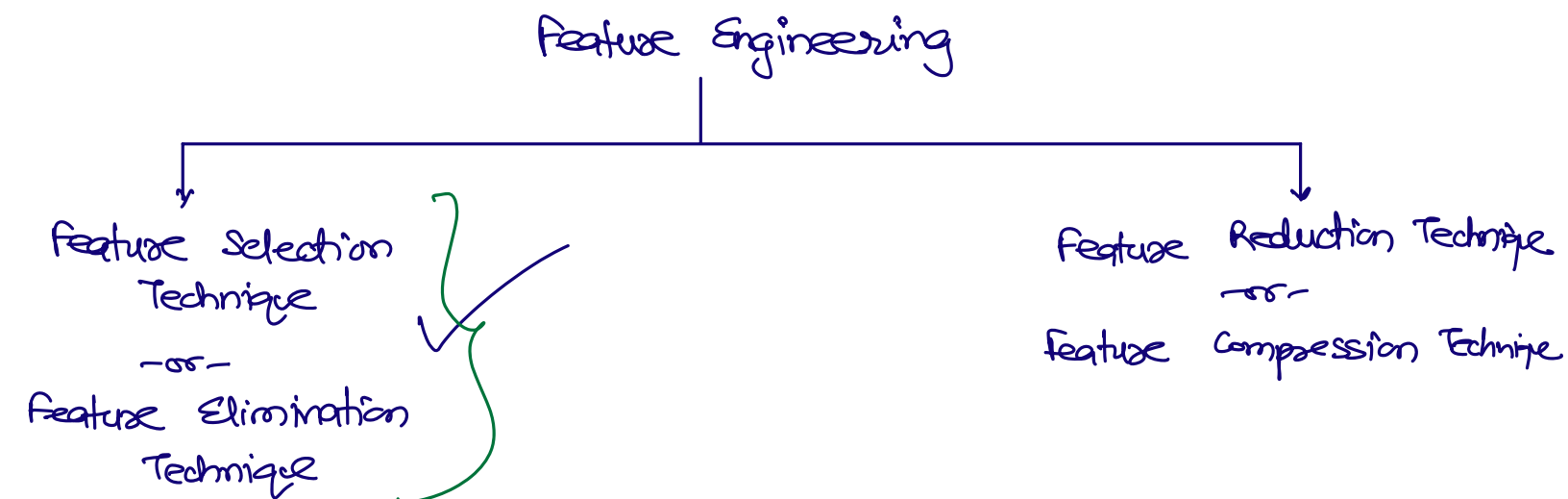
SELECT/SHORTLIST the set of **SIGNIFICANT FEATURES**

STATISTICALLY SIGNIFICANT FEATURES

based on:

- ① Label column → correlation analysis
 - ② ML algorithm → RFE, SPI
 - ③ **Statistical Testing formula.** if feature cols is Normally Dist.:
 - Parametric Test ✓
 - else:
 - Non-parametric Test ✓
- OLS

so that you can ACHIEVE AN OPTIMAL MODEL.



Classification

features: any kind of data
label: categorical / discrete ND.

Binary Classification

Label column will have only two unique values.

eg: 0 | 1
true | false
ham | spam

multi-class Classification

Label column will have more than two unique values.

eg: Primary | Social | Promotions | Updates

Logistic Regression

Applying sigmoid function on regression line

$$\text{Sigmoid } f^{\pi} = \frac{1}{1 + e^{-y}}$$

... returns probability
0 to 1

where,

In logistic regression

y = equation of line.

Standard threshold

$0 - 0.5 \rightarrow 0$

$> 0.5 \rightarrow 1$